

SYSTEM SECURITY AND PRIVACY PLAN (SSPP)

FOR THE EXPEDITIONARY COMMAND NODE (ECN-100)

System Identifier: ECN-100

Prepared For: Supported Operational Unit (representative/fictional)

Prepared By: TechCoaches Information Systems Design

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This describes an entirely hypothetical system, organization, and equipment set, created for illustrative purposes; it is not based on and does not represent any actual fielded platform, program of record, or unit.

Provided as a technical writing sample — Nathan B. Smith.

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RECORD OF CHANGES

Version	Date	Description of Change	Author
1.0	17 Aug 2026	Initial issue	TechCoaches Information Systems Design

SAMPLE

TABLE OF CONTENTS

- 1.0 Introduction
 - 1.1 Purpose
 - 1.2 Scope and Applicability
 - 1.3 References
- 2.0 System Identification
 - 2.1 System Name, Mission, and Ownership
 - 2.2 Responsible Personnel
 - 2.3 Security Categorization
- 3.0 System Environment and Authorization Boundary
- 4.0 Interconnections
- 5.0 Security Control Implementation
- 6.0 Privacy Overlay
- 7.0 Plan of Action and Milestones (POA&M)
- Appendix A — Ports, Protocols, and Services (PPS) Registration
- Appendix B — Hardware and Software Baseline Inventory
- Appendix C — Control Implementation Procedures
- Appendix D — Diagrams

1.0 INTRODUCTION

1.1 Purpose

This System Security and Privacy Plan (SSPP) documents the security and privacy controls implemented for the Expeditionary Command Node (ECN-100), in support of an Assessment and Authorization (A&A) decision under the DoD Risk Management Framework (RMF), per DoDI 8510.01. Controls are drawn from NIST SP 800-53, with impact levels assigned per CNSSI No. 1253.

In plain terms: this document tells the Authorizing Official (AO) what the system is, what risks it carries, and what has been done to manage those risks, so the AO can decide whether to grant it permission to operate.

1.2 Scope and Applicability

This SSPP applies to all hardware, software, and firmware physically installed within the ECN-100 shelter/trailer and its organic radio and satellite communications (SATCOM) equipment. It does not cover the networks the ECN-100 connects to once established in theater; those are addressed under Section 4.0, Interconnections, and governed individually by Interconnection Security Agreements (ISAs).

1.3 References

- a. DoD Instruction 8510.01, Risk Management Framework (RMF) for DoD Systems
- b. NIST Special Publication 800-53, Security and Privacy Controls for Information Systems and Organizations
- c. Committee on National Security Systems Instruction (CNSSI) No. 1253, Security Categorization and Control Selection for National Security Systems
- d. NIST Special Publication 800-37, Risk Management Framework for Information Systems and Organizations

2.0 SYSTEM IDENTIFICATION

2.1 System Name, Mission, and Ownership

The Expeditionary Command Node (ECN-100) is a trailer- and shelter-mounted mobile command and control (C2) platform. It houses a self-contained tactical local area network, SATCOM terminal, and line-of-sight (LOS) radio relay equipment, and operates on generator power, vehicle power, or shore power depending on the deployment scenario.

Unlike a fixed facility, the ECN-100's authorization boundary moves with the platform. It is designed to stand up, operate, and displace within a single operational period, which drives several of the control decisions in Section 5.0 — particularly physical security during transit and continuity of operations when connectivity is lost.

The ECN-100 carries two connected enclaves: SIPRNET (classified, Secret) and NIPRNET (unclassified but sensitive). Both enclaves route over the same physical SATCOM uplink but remain logically and cryptographically separated end to end. A separate, physically isolated NIPRNET segment is also present for local administrative use; because it is air-gapped, it has no external connectivity and is not part of either enclave's interconnection path.

Primary connectivity for both connected enclaves is via the FTX-2400 FTSAT. Should the FTSAT be unavailable — due to obstruction, maintenance, or displacement to a new location — the ECN-100 falls back to alternate connectivity over high-speed civilian infrastructure (commercial broadband or cellular data service where available), used strictly as an unencrypted bearer for traffic that is already encrypted before it reaches that bearer.

Navy message traffic (fleet broadcast) is received independently of the FTSAT data path, primarily via UHF SATCOM with HF as the alternate receive path.

Workstations. The permanent workstation baseline consists entirely of laptops: four GCCS-M workstations on SIPRNET processing OTH-T Gold Red/White/Blue track data for contact reporting, one SIPRNET administration workstation, and one dedicated NIPRNET workstation. Additional workstations can be added to either enclave as mission requirements warrant.

The ECN-100's SATCOM suite consists of two terminals. An INMARSAT B terminal provides a lower-data-rate commercial backup path. An FTX-2400 Flyaway Tri-Band SATCOM Terminal (FTSAT) — a Commercial-Off-The-Shelf (COTS), 2.4-meter transportable terminal operating across C-, X-, and Ku-band — serves as the primary high-data-rate uplink, compatible with DSCS, NATO, INTELSAT, EUTELSAT, PANAMSAT, and DOMSAT satellite systems, and capable of hub-independent set-up, acquisition, and operation without depending on a specific ground station configuration.

The ECN-100 provides a deployable C2 capability for a supported unit operating away from garrison infrastructure: voice and data connectivity back to higher headquarters, local situational awareness display, and short-term data storage for mission-essential functions.

System Owner: TechCoaches Information Systems Design (fictional prime)

Supported Organization: Representative U.S. Navy operational unit

This exercise describes an entirely hypothetical system and organization. It does not represent, and is not based on, any actual fielded platform, program of record, or unit. All equipment names and designators in this document — including ECN-100, FTX-2400, UST-54X, and FBMP-200 — are invented for this exercise and do not refer to real hardware.

2.2 Responsible Personnel

Role	Responsibility
Authorizing Official (AO)	Accepts residual risk and grants authorization to operate
Information System Security Manager (ISSM)	Oversees the system's security program
Information System Security Officer (ISSO)	Day-to-day security operations and this SSPP's maintenance

2.3 Security Categorization

Representative categorization for this exercise, per CNSSI No. 1253:

Confidentiality	Integrity	Availability
Moderate	Moderate	High

Availability is elevated above the other two objectives because loss of the C2 picture during an active operation carries a materially different impact than a comparable outage at a garrison facility.

3.0 SYSTEM ENVIRONMENT AND AUTHORIZATION BOUNDARY

The authorization boundary encompasses the tactical LAN, C2 server, crew workstations, local identity store, and SATCOM/LOS radio equipment physically housed within the ECN-100. A system/network architecture diagram, a physical equipment layout, a network topology diagram, and a data flow diagram are provided in Appendix D.

4.0 INTERCONNECTIONS

The ECN-100 maintains two external network enclaves, each governed by its own Interconnection Security Agreement (ISA), executed prior to connection and reviewed at each reauthorization cycle. Both enclaves share the same physical uplink hardware but never share a cryptographic boundary — SIPRNET traffic is protected by Type 1 encryption regardless of which physical bearer carries it, which is what makes the civilian-infrastructure fallback acceptable as an alternate path rather than a separate risk decision.

Enclave	Primary Path	Alternate Path	Encryption	ISA
SIPRNET (classified, Secret)	FTX-2400 FTSAT organic uplink	Civilian infrastructure — unencrypted bearer only	NSA-certified Type 1 in-line encryptor; ciphertext before either bearer	ISA-SIPR-01
NIPRNET (unclassified, sensitive)	FTX-2400 FTSAT organic uplink	Civilian infrastructure	Garrison-standard IPsec VPN encryptor	ISA-NIPR-01
Navy Fleet Broadcast (message traffic)	UHF SATCOM transceiver	HF radio system	Military COMSEC equipment (representative)	ISA-FLTBCST-01

A separate, physically air-gapped NIPRNET segment is present for local administrative use. It is intentionally excluded from the table above: because it has no external connectivity, it is not an interconnection and carries no ISA.

No cross-domain solution is in scope for this representative system.

5.0 SECURITY CONTROL IMPLEMENTATION

The table below summarizes implementation status by NIST SP 800-53 control family, with mobile-platform-specific notes. Detailed procedures for the Access Control and Incident Response families are provided in Appendix C.

Family	What It Covers	Status	Mobile-Specific Note
AC	Who/what can log on and what they can touch	Implemented	Local identity store; see Appendix C
AT	Security training for the crew	Implemented	Delivered pre-deployment; no in-theater refresh assumed
AU	What gets logged and reviewed	Implemented	Local buffering, scheduled offload at reconnection
CM	Keeping the configuration baseline stable	Implemented	Baseline frozen at departure; changes tracked in POA&M
CP	Staying operational through a disruption	Implemented	DDIL operating mode; see Appendix C
IA	Proving you are who you say you are	Implemented	PKI/CAC with offline fallback
IR	Handling a security incident	Implemented	Local containment; reachback reporting on reconnect
MA	Maintaining equipment without introducing risk	Implemented	Field maintenance kit is boundary-scoped and

Family	What It Covers	Status	Mobile-Specific Note
			inventoried
MP	Protecting removable media	Implemented	Field sanitization procedure required before reuse
PE	Physical access to the equipment	Implemented	Covers transit/convoy posture and EMCON
PS	Vetting the people who operate the system	Implemented	Clearance verification tied to crew manifest
RA	What could go wrong and how bad it would be	Implemented	Includes loss/capture of equipment as a scenario
SC	Protecting data on the wire and over RF/SATCOM	Implemented	SIPRNET Type 1, NIPRNET IPsec VPN; see Section 4.0
SI	Keeping software patched and behaving as expected	Partial	Patch cadence limited by intermittent connectivity; see POA&M

6.0 PRIVACY OVERLAY

The ECN-100 processes limited personnel accountability data (crew manifest, credential status) but no broader PII/PHI dataset. A Privacy Impact Assessment (PIA) and System of Records Notice (SORN) applicability review are not required for this representative system given that limited scope.

7.0 PLAN OF ACTION AND MILESTONES (POA&M)

Item	Control	Finding	Remediation	Target Date
1	SI-2	Patch cadence constrained by intermittent SATCOM connectivity	Establish scheduled patch-pull window at each reconnection	TBD

APPENDIX A — PORTS, PROTOCOLS, AND SERVICES (PPS) REGISTRATION

Service	Port/Protocol	Purpose	Boundary
C2 chat/messaging	TCP/5222	Text-based command and control coordination	Internal + SATCOM uplink
Situational awareness feed	UDP/4500	Track/position data distribution to displays	Internal only
Directory/authentication sync	TCP/636 (LDAPS)	Identity store replication at reconnection	SATCOM uplink, scheduled window
Network time sync	UDP/123 (NTP)	Time source for logging and crypto sync	Internal, local GPS-disciplined source
Remote system management	TCP/22 (SSH)	Administrative access to network appliances	Internal only, key-based auth

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APPENDIX B — HARDWARE AND SOFTWARE BASELINE INVENTORY

Component	Type	Version/Model	Role
Tactical LAN switch	Hardware	Representative 24-port ruggedized switch	Internal network backbone
SIPRNET router/switch	Hardware	Representative ruggedized router/switch pair	Classified enclave LAN
NIPRNET router	Hardware	Commercial-grade router (representative)	Connected unclassified enclave LAN
Air-gapped NIPRNET router	Hardware	Commercial-grade router (representative, lower-capability)	Isolated administrative segment; no external connectivity
SIPRNET encryption device	Hardware	NSA-certified Type 1 in-line network encryptor (representative)	Protects classified traffic before any physical bearer
NIPRNET encryption device	Hardware	Garrison-standard IPsec VPN encryptor (representative)	Protects unclassified/sensitive traffic in transit
GCCS-M/OTH-T Gold workstations (4)	Hardware/Software	Ruggedized laptops, hardened OS baseline, GCCS-M client	SIPRNET; Red/White/Blue track contact reporting
SIPRNET administration workstation	Hardware/Software	Ruggedized laptop, hardened OS baseline	SIPRNET; system/network administration
NIPRNET workstation	Hardware/Software	Ruggedized laptop, hardened OS baseline	Connected NIPRNET; unclassified operator use
Crew workstations	Hardware/Software	Ruggedized laptops, hardened OS baseline	General operator terminals; additional units as warranted
INMARSAT terminal	Hardware	INMARSAT B, case-mounted, with crypto/power/UPS	External SATCOM uplink (commercial backup path)
Flyaway Tri-Band SATCOM Terminal (FTSAT)	Hardware	FTX-2400, COTS 2.4-meter tri-band (C/X/Ku)	External SATCOM uplink (primary high-data-rate path)
Alternate connectivity modem	Hardware	Commercial broadband/cellular data modem	Alternate bearer; carries pre-encrypted traffic only
UHF SATCOM transceiver	Hardware	UST-54X (representative)	Primary receive path for Navy message traffic
HF radio system	Hardware	Representative dual HF transceiver set	Alternate receive path for Navy message traffic
Message processing system	Hardware/Software	FBMP-200 (representative)	Processes and routes fleet broadcast message traffic
C2 server	Hardware/Software	Rack-mount server, hardened OS baseline	Hosts C2 and identity applications
Local identity store	Software	Replicated directory service instance	Offline-capable authentication
COMSEC key fill device	Hardware	Representative key loader	Cryptographic key

Component	Type	Version/Model	Role
			management

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APPENDIX C — CONTROL IMPLEMENTATION PROCEDURES

C.1 Implementing Access Control for the ECN-100 Network

Performed during pre-deployment configuration, before the ECN-100 leaves garrison.

1. Provision PKI/CAC-based authentication against a local, on-platform identity store, replicated from garrison before departure so the ECN-100 does not depend on a live connection back to a garrison domain controller.
2. Configure role-based access control (RBAC) groups for each crew position (watch officer, communications operator, system administrator).
3. Configure an offline/local authentication fallback for use during disconnected, intermittent, or limited-bandwidth (DDIL) operations. Fallback credentials are single-use and time-limited, and every use is logged for review at the next reconnection.
4. Disable or remove any default or vendor accounts on installed equipment.
5. Document any control not fully implementable in the mobile environment, and the compensating measure in place, in the POA&M (Section 7.0).

C.2 Executing Incident Response in a DDIL Environment

DDIL — disconnected, intermittent, or limited-bandwidth — is the operating condition the ECN-100 is built around. This procedure assumes the platform cannot call back to a garrison Security Operations Center (SOC) at the moment an incident is detected.

1. Detect and isolate the affected system or segment from the rest of the ECN-100 LAN.
2. Contain locally using on-platform tools; do not wait for reachback connectivity.
3. Log all actions taken, with timestamps, to the local audit store. Local logging is mandatory because the audit trail may be the only record available until a garrison SOC begins its own review.
4. Continue mission-essential operations on unaffected systems where possible.
5. Upon reconnection, report the incident and transmit local logs to the garrison SOC per the unit's incident reporting timeline.
6. Participate in an after-action review and update local procedures or the POA&M as needed.

APPENDIX D — DIAGRAMS

The five diagrams below support the authorization boundary described in Section 3.0 and the interconnections described in Section 4.0. Each is rendered in monochrome, hand-authored as SVG, and built to standard Visio-stencil conventions (network diagram, floor plan, detailed network diagram, data flow diagram, and cross-functional flowchart respectively) so it reproduces in Microsoft Visio without a redesign.

ECN-100 System / Network Architecture Diagram

Hypothetical system — for illustrative purposes only. No CUI or classified information.

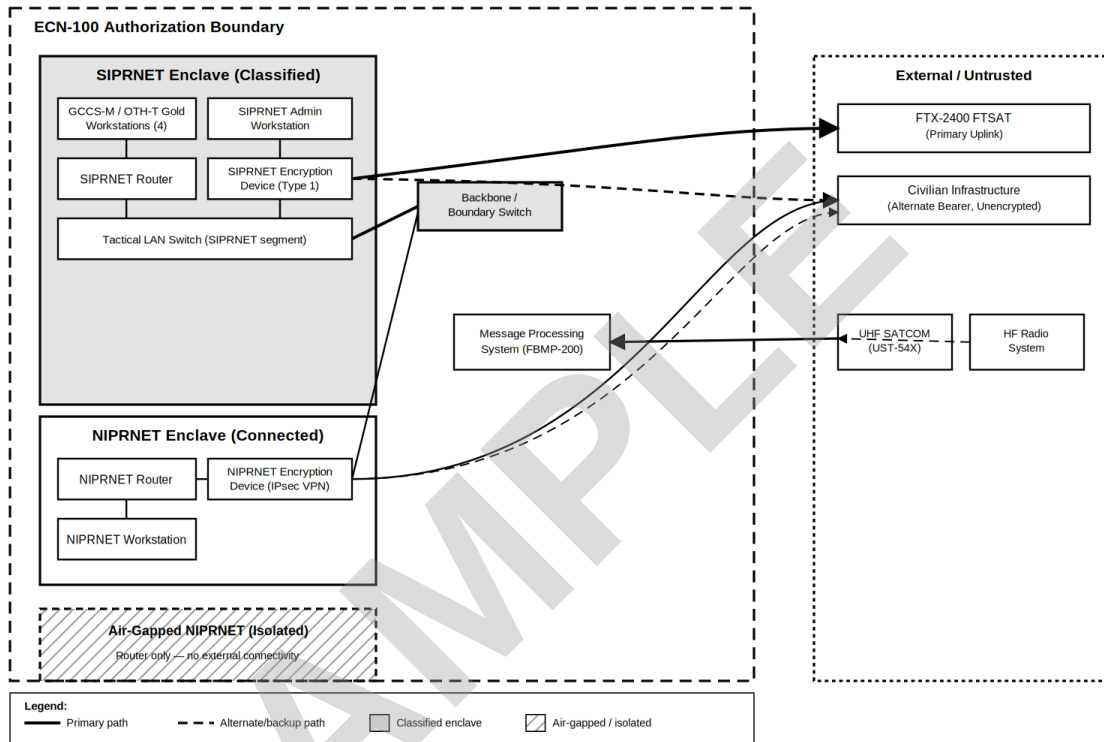


Figure D-1. System / Network Architecture Diagram

ECN-100 Physical Equipment Layout

Hypothetical system — for illustrative purposes only. Representative layout, not to scale.

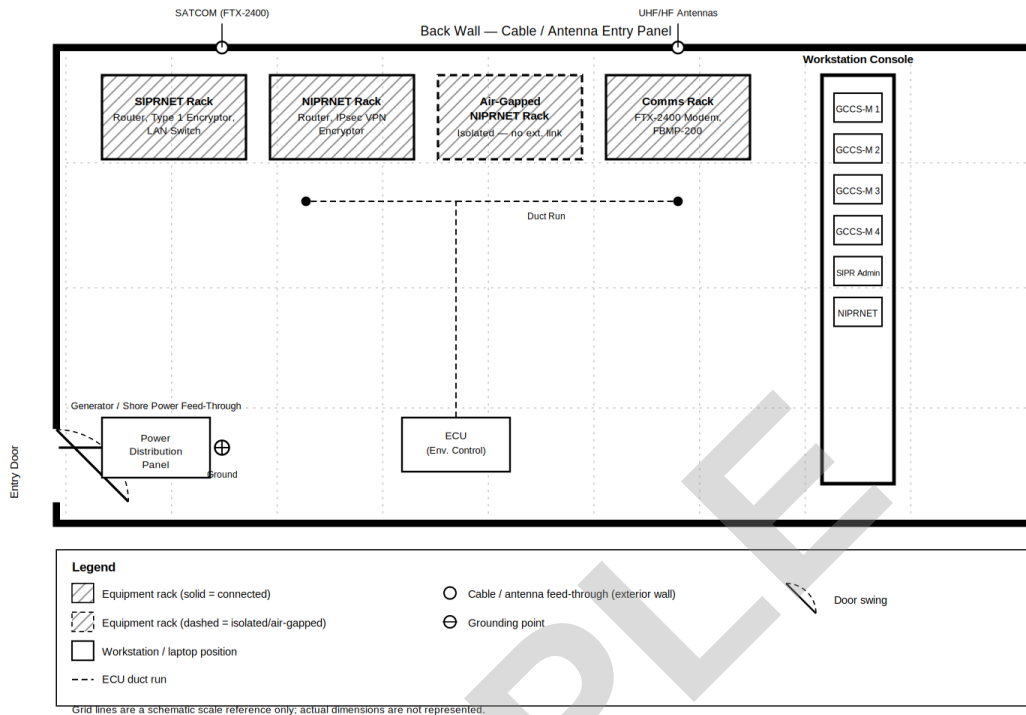


Figure D-2. Physical Equipment Layout

ECN-100 Network Topology Diagram

Hypothetical system — for illustrative purposes only. Port labels are representative.

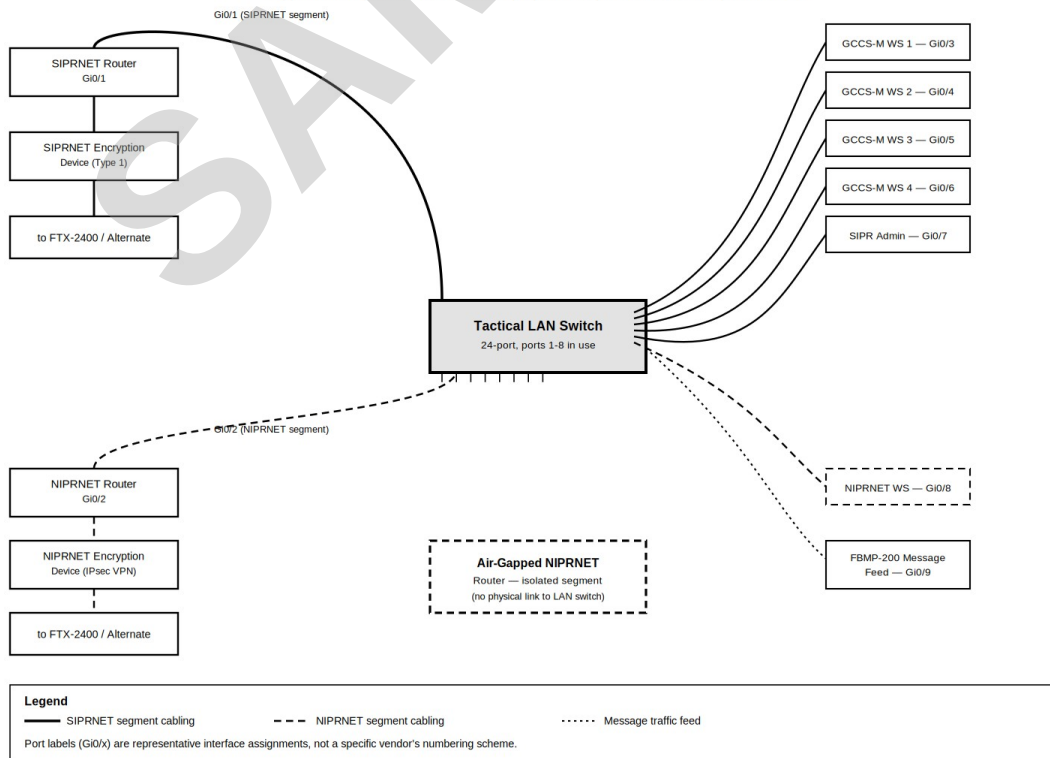


Figure D-3. Network Topology Diagram

ECN-100 Data Flow Diagram

Hypothetical system — for illustrative purposes only. Level 1 DFD, trust boundary crossings marked.

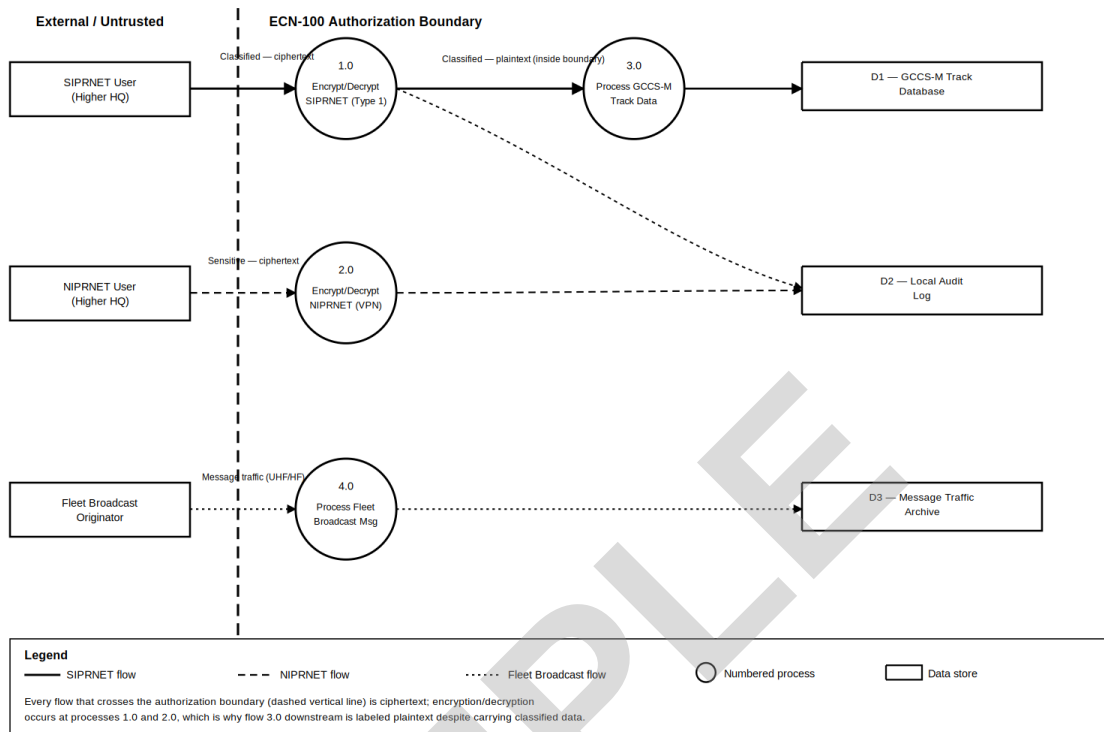


Figure D-4. Data Flow Diagram

ECN-100 Incident Response Workflow (DDIL)

Hypothetical system — for illustrative purposes only. Cross-functional swimlane.

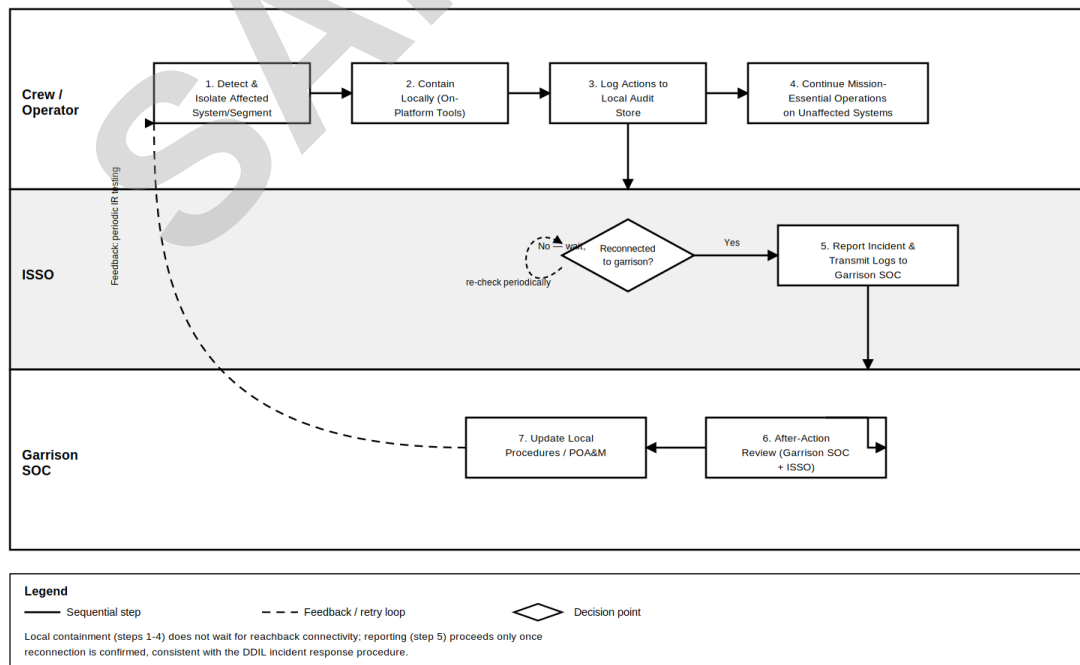


Figure D-5. Incident Response Workflow (DDIL)

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